Lesson 9: Project: My Simulation

Module 3 | 50 minutes | Project: simulation | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. assumption ___ A. A value recorded to answer the question.
b. controlled experiment ___ B. Changing one thing and comparing.
c. measurement ___ C. Something the model treats as true or leaves out.

2. Rule, assumption or measurement?

Write each item in the right group: Rule / Assumption / Measurement.

a. Each fish moves 3 steps in a random direction → _____
b. The pond has no predators → _____
c. Count fish at the end of each day → _____
d. If a shark touches a fish, the fish disappears → _____
e. All fish swim at the same speed → _____

3. My plan

Plan your project: Simulate a real-world process to answer a question: write rules and assumptions, build it with randomness and agents or time steps, measure a result, and run an experiment by changing one setting. Sketch it and list what each part will do.

4. Checklist

Tick each one when your project does it.

a. ☐ Question
b. ☐ Rules and assumptions
c. ☐ Randomness
d. ☐ Agents or time steps
e. ☐ Measurement
f. ☐ Experiment and conclusion

5. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ My simulation has rules, randomness and a measurement.
☐ I ran an experiment by changing one setting.
☐ I can explain my model's assumptions.